

Introduction to Computer Programming (ICP, CCIT4020)

HKU SPACE Community College, 2021-2022 Semester 1

(Lab 09)

(Practicing Lab, NOT for submission)

Review the related lecture note(s), and finish exercises below.

1. Writing Text Data to Text File:

Write a Python program (named **Lab9_1.py**) with function **main()** to 1) let user input a file name (e.g. "CL02.txt") for creating a data file; 2) then properly call the **writedata()** function to write **3** student records (Name and Mark for Programming) to the data file as below:

- Also write the function **writedata()** in the same program file, which:
 - Input 3 records and store in variables named `studName` and `icp` in string types
 - Write 3 records to the data file by `write()` (store line by line in the data file).

(Hint: for loop is used in the **writedata()**)

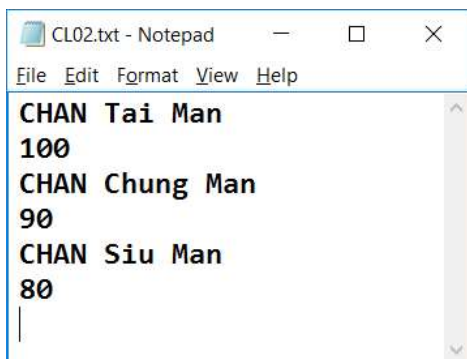
Sample program output is shown as below:

```
Enter a OUTPUT file name for writing: CL02.txt
Enter record for 1 [1.Name and 2.Mark:]
Student name: CHAN Tai Man
Mark for ICP: 100
Enter record for 2 [1.Name and 2.Mark:]
Student name: CHAN Chung Man
Mark for ICP: 90
Enter record for 3 [1.Name and 2.Mark:]
Student name: CHAN Siu Man
Mark for ICP: 80
OUTPUT file writing finished OK!
```

User
Input

* Check if there is a file created as expected (and its content)

Sample Output File Created as below:



```
CL02.txt - Notepad
File Edit Format View Help
CHAN Tai Man
100
CHAN Chung Man
90
CHAN Siu Man
80
```

* IMPORTANT: the file must be closed by `close()` to disconnect both the program and the file.

* Remark: File existence checking is needed in **main()** to check the file in the storage medium.

2. **Reading Text Data from Text File: (Use the data file created in Question 1):**

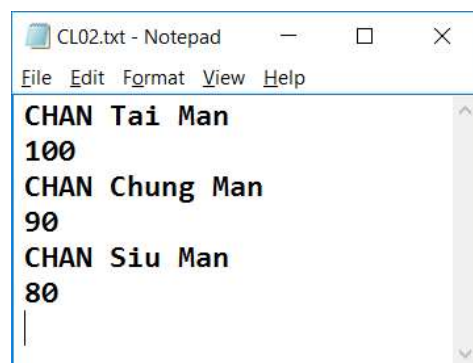
Write a Python program (named `Lab9_2.py`) with function `main()` to 1) let user input a file name (e.g. `"CL02.txt"`) **for opening the data file created from Question 1**; 2) then properly call the `readdata()` function to read student records (Student Name and Mark for Programming ICP) from the data file below:

- Also write a function `readdata()` in the same program file, which:
 - Read students' records and store in variables: `stud` (in string type) and `icpr` (in integer type).
 - Determine the average of ICP marks.
 - Display students' records (Student name and Mark for ICP) (See sample program output)
 - Display the average with the decimal point for 1 digit at the end.

(Hint: may use function `readlines()` to read all text file content as string list, or function `read()` to read all text file content as one string, in `readdata()`)

* IMPORTANT: the file must be closed by `close()` to disconnect both the program and the file.

* Remark: File existence checking is needed in `main()` to check the file in the storage medium



Sample Input File:

Sample program output is shown as below:

```
Enter a INPUT file name for reading CL02.txt
CL02.txt exists in the current directory
The student records include:
-Student Name: CHAN Tai Man
  Mark for ICP: 100
-Student Name: CHAN Chung Man
  Mark for ICP: 90
-Student Name: CHAN Siu Man
  Mark for ICP: 80
The average of ICP for 3 students is 90.0
```

3. *** Optional ***

Writing data to the binary file, and read them back with display:

Write a Python program (named `Lab9_3.py`) with function `main()` to

- 1) let user input a file name (e.g. `"data.bin"`) for creating a data file;
- 2) properly call the `writeBinData()` function to write **30 random numbers** within a range of one byte (0~255) to the binary file, finally
- 3) properly call the `readBinData()` function to read data back from the binary file, and display them back:

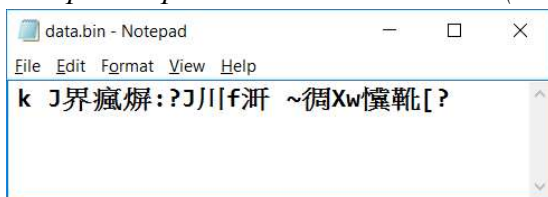
- Also write the functions `writeBinData()` and `readBinData()` in the same program file to achieve the task.

Sample program output is shown as below:

```
Enter an OUTPUT BINARY file name for writing: data.bin
OUTPUT BINARY file writing finished OK! Data below:
[107, 17, 74, 172, 201, 186, 198, 152, 98, 58, 169, 153,
74, 164, 116, 102, 210, 84, 22, 126, 213, 80, 88, 119, 242,
65, 185, 117, 91, 169]
```

```
The binary data READ OK, and data (in a form of List) are:
[107, 17, 74, 172, 201, 186, 198, 152, 98, 58, 169, 153,
74, 164, 116, 102, 210, 84, 22, 126, 213, 80, 88, 119, 242,
65, 185, 117, 91, 169]
```

Sample Output File Created as below (when opened with text editor such as notepad):



~ END ~